

Unit 6

ELECTROCHEMISTRY

47 items · 97 marks

Syllabus sections in this unit

6.1 Redox processes: electron transfer and changes in oxidation number (oxidation state)

Unit 6: Electrochemistry

1. M/J 26/P21/Q3/d

- 3 Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄.
Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

(d) Complete Table 3.2.

Table 3.2

species	oxidation number of sulfur
H ₂ SO ₄	
SO ₂	
H ₂ S	

[2]

2. M/J 26/P21/Q4/a(ii)

- 4 (a) Compound **Z** is oxidised to produce organic compound **W**, C₅H₆O₇, by heating under reflux with an excess of acidified Cr₂O₇²⁻(aq).

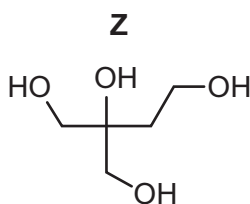
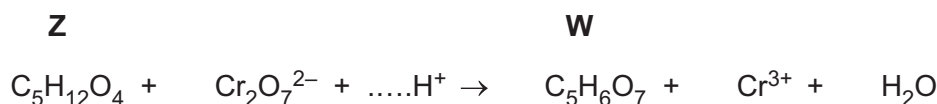


Figure 4.1

- (ii) The complete oxidation of 0.025 mol of **Z** requires 0.050 mol of Cr₂O₇²⁻(aq).
W is produced in this reaction.

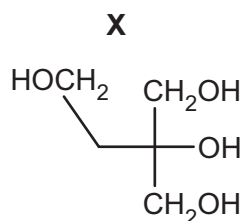
Balance the equation for this reaction.



[1]

3. M/J 26/P22/Q4/c

also Unit 17 (2 marks); this unit 2 marks

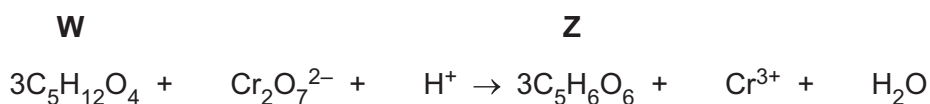
(b) The structure of compound **X** is shown in Figure 4.2.**Figure 4.2**

The molecular formula of **X** is $\text{C}_5\text{H}_{12}\text{O}_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of 0.500 mol dm^{-3} acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$.

(c) (i) Compounds **X** and **W** are a pair of structural isomers.

0.0012 mol of **W** is completely oxidised when it is heated under reflux with exactly 0.0020 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce compound **Z**.

Complete the equation for this reaction.



[2]

(ii) Each molecule of **W** contains four hydroxyl groups. Only two of these are primary hydroxyl groups.

Table 4.1 shows the observations when separate samples of **Z** are added to each reagent.

Table 4.1

reagent	observations with Z
2,4-dinitrophenylhydrazine (2,4-DNPH)	orange precipitate
Fehling's reagent	none

Use the information in **4(c)** to suggest the structure of **W**.



[2]

4. M/J 26/P23/Q4/d(ii)

4 The structure of compound **V** is shown in Figure 4.1.

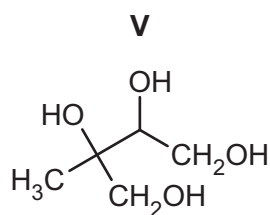


Figure 4.1

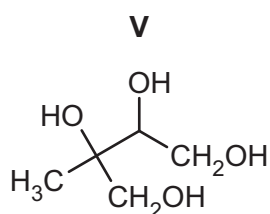


Figure 4.1

(c) Compounds **W**, **X** and **Y** are structural isomers of **V**.

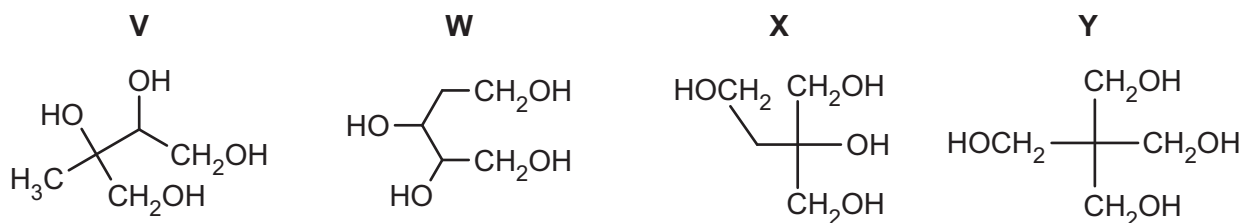
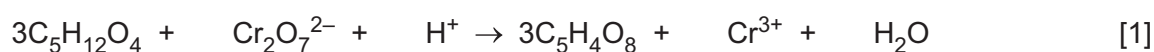


Figure 4.2

(d) Exactly 0.03 mol of one of the isomers, **V**, **W**, **X** or **Y**, is completely oxidised by 0.08 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce only **one** organic compound with molecular formula $\text{C}_5\text{H}_4\text{O}_8$.

(ii) Complete the equation for this reaction.



5. M/J 25/P21/Q4/b(iii)

4 The structure of vitamin C is shown in Fig. 4.1.

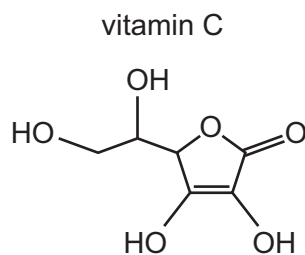


Fig. 4.1

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

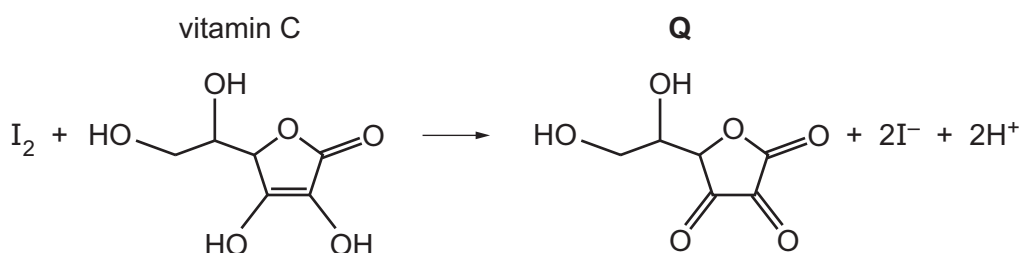


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm^3 of vitamin C solution. 5.00 cm^3 of this vitamin C solution is added to a flask with approximately 150 cm^3 of water and an indicator. Exactly 28.40 cm^3 of $5.00 \times 10^{-4}\text{ mol dm}^{-3}$ $\text{I}_2(\text{aq})$ reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(iii) Deduce the role of $\text{I}_2(\text{aq})$ in the reaction in Fig. 4.2.

[1]

6. M/J 25/P24/Q2/b(ii)

(b) SCl_2 is a red liquid that reacts with water.



(ii) When SCl_2 reacts with water, the SCl_2 is broken down and a disproportionation reaction occurs.

State the oxidation numbers of sulfur in SCl_2 and H_2SO_3 .

oxidation number of S in SCl_2

oxidation number of S in H_2SO_3

[2]

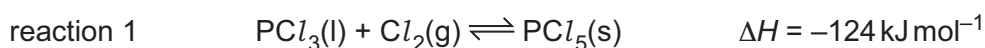
7. MAR 25/P22/Q1/a

also Unit 7 (2 marks), Unit 2 (2 marks), Unit 4 (2 marks); this unit 2 marks

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.**(a)** Chlorine forms three different compounds with phosphorus.The most common compounds are PCl_3 and PCl_5 .**(i)** Complete Table 1.1.**Table 1.1**

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(ii) In a closed system, PCl_3 and PCl_5 exist in an equilibrium mixture as shown in reaction 1.Deduce **two** conditions that favour the production of PCl_5 in reaction 1.

1

2

[2]

(iii) The third compound of phosphorus and chlorine, **W**, has a relative molecular mass, M_r , between that of PCl_3 and PCl_5 . The compound contains 69.6% by mass of chlorine.Determine the molecular formula of **W**.molecular formula of **W** =

[2]

(iv) **W** is a liquid at room temperature and pressure. It reacts vigorously with water to form an acidic solution.Suggest the structure and bonding in **W**. Explain your answer.

[2]

8. O/N 24/P21/Q2/c(ii,iii)**2** Chlorine is one of the elements in Group 17 of the Periodic Table.**(c)** CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO . NaClO is made from chlorine in a disproportionation reaction.**(ii)** Define disproportionation.

[1]

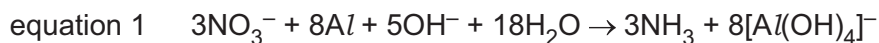
(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaClO .

[1]

9. O/N 24/P22/Q3/b(i,ii)

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(b) A common test for nitrates is the reaction with NaOH and Al. Equation 1 shows the reaction.



(i) Deduce the oxidation state of nitrogen in NO_3^- .

[1]

(ii) Identify the species that is oxidised in equation 1.

[1]

10. O/N 24/P23/Q2/c(ii,iii)

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO.

NaClO is made from chlorine in a disproportionation reaction.

(ii) Define disproportionation.

[1]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaClO.



[1]

11. M/J 24/P23/Q1/d(i)

(d) The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate, CaSiO_3 . The oxidation number of calcium in CaSiO_3 is +II.

(i) Deduce the oxidation number of silicon in calcium silicate.

oxidation number of silicon

[1]

12. MAR 24/P22/Q1/b(i),c,d(i)

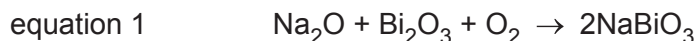
1 Bismuth is an element in Group 15 of the Periodic Table.

(b) Bismuth reduces water to form bismuth oxide, Bi_2O_3 . A colourless gas that ignites with a squeaky pop also forms.

(i) Construct an equation for the reduction of water by bismuth.

[1]

(c) Bi_2O_3 can be used to form NaBiO_3 , as shown in equation 1.



(i) Deduce the oxidation number of Bi in Bi_2O_3 and in NaBiO_3 .

oxidation number of Bi:

in Bi_2O_3

in NaBiO_3

[1]

(ii) Identify the reducing agent in equation 1.

[1]

(d) NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

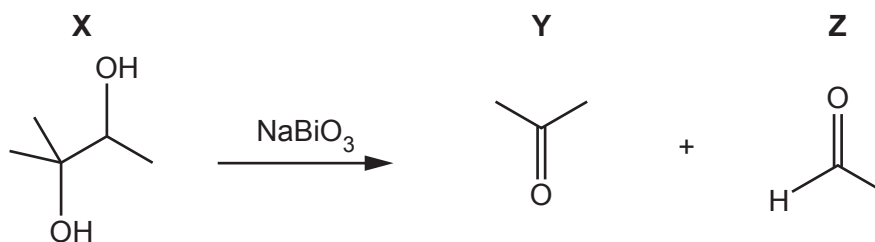


Fig. 1.1

(i) Explain the term oxidising agent.

[1]

13. O/N 23/P21/Q3/a,c

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(a) Define oxidising agent.

[1]

(c) When KClO_3 is heated without a catalyst, KClO_4 and KCl form.



Explain why this reaction is described as a disproportionation reaction.

[1]

14. O/N 23/P22/Q3/b(iii)

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

(b) H_3PO_4 is also formed in the process shown in reaction 1.

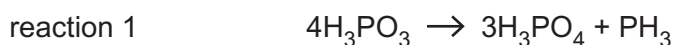


Table 3.1 shows some relevant thermodynamic data.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
H_3PO_3	-972
H_3PO_4	-1281
PH_3	+9

(iii) Explain why reaction 1 is a disproportionation reaction.

Explain your reasoning with reference to relevant oxidation numbers.

[2]

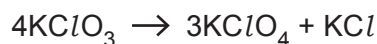
15. O/N 23/P23/Q3/a,c

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(a) Define oxidising agent.

[1]

(c) When KClO_3 is heated without a catalyst, KClO_4 and KCl form.



Explain why this reaction is described as a disproportionation reaction.

[1]

16. M/J 23/P22/Q2/e(ii)

2 Chlorine is a reactive element. It forms many compounds.

(e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaClO , to water.

(ii) Sodium chlorate(I), NaClO , reacts with hydrogen peroxide to produce sodium chloride, water and oxygen gas.

Construct an equation for this reaction.

[1]

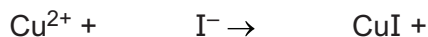
17. M/J 23/P23/Q1/d(i,ii)

1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

- (d) When KI(aq) is added to CuSO₄(aq) the blue-coloured solution turns brown and a white precipitate of CuI(s) is seen.

The reaction between copper ions and iodide forms only two products.

- (i) Complete the equation for this reaction.



[1]

- (ii) Identify the oxidising agent in this reaction. Explain your answer in terms of electron transfer.

[1]

18. M/J 22/P22/Q1/b(i,ii)

- (b) When magnesium is heated in air, magnesium oxide, MgO, is the major product. Smaller amounts of magnesium nitride, Mg₃N₂, are also made.

- (i) Calculate the oxidation number for magnesium and for the nitrogen species in Mg₃N₂ to complete Table 1.1.

Table 1.1

species	magnesium in Mg ₃ N ₂	nitrogen in Mg ₃ N ₂
oxidation number		

[1]

- (ii) Identify the type of reaction which takes place between magnesium and nitrogen. Explain your answer.

[1]

19. MAR 22/P22/Q2/a(i),c(ii)

2 Some oxides of elements in Period 3 are shown.



- (a) Na reacts with O₂ to form Na₂O. Na is the reducing agent in this reaction.

- (i) Define reducing agent.

[1]

- (c) P₄O₆ is a white solid that has a melting point of 24 °C. Solid P₄O₆ reacts with water to form H₃PO₃.

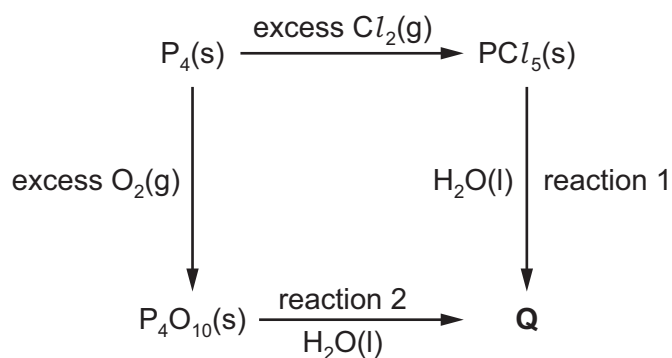
- (ii) Determine the oxidation number of P in H₃PO₃.

[1]

20. O/N 21/P21/Q3/b(i)

3 Phosphorus is a reactive Period 3 element.

(b) Some reactions of $P_4(s)$ are shown in the reaction scheme.



(i) State the oxidation number of phosphorus in P_4O_{10} .

[1]

21. O/N 21/P22/Q1/d

1 Hydrogen iodide, HI, is a colourless gas at room temperature.

(d) HI reacts with oxygen to form iodine and water.

(i) Construct an equation for the reaction of HI with oxygen.

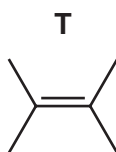
[1]

(ii) Explain, with reference to oxidation numbers, why this reaction is a redox reaction.

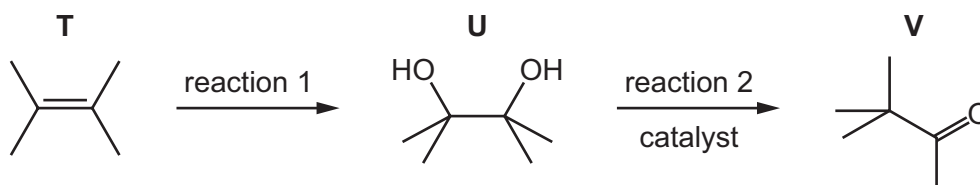
[2]

22. O/N 21/P22/Q3/e(ii)

3 Compound **T** is an isomer of C_6H_{12} .

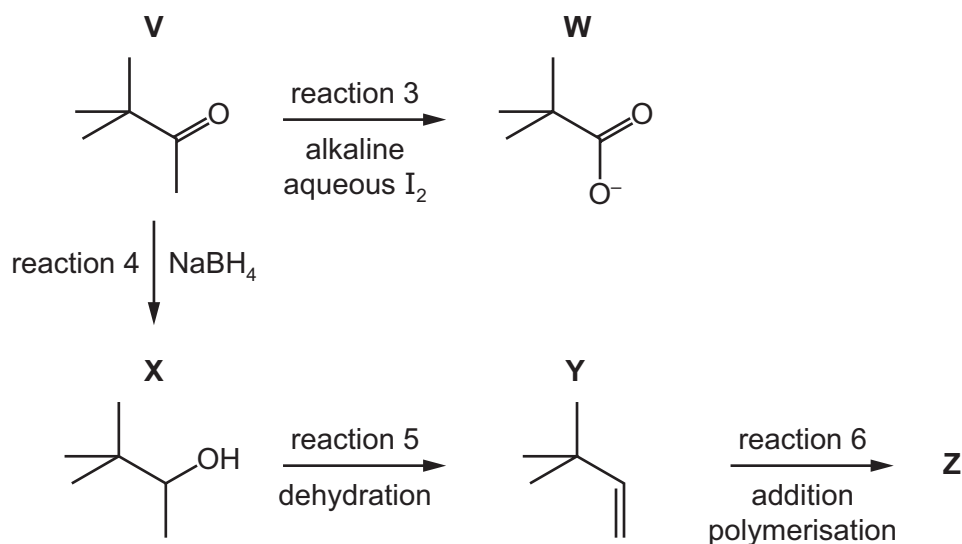


(d) A reaction scheme starting with **T** is shown. Reaction 2 occurs in the presence of a catalyst; knowledge of the mechanism for this reaction is not required.



(e) **V** is used in a wide range of organic reactions.

Some reactions of **V** are shown.



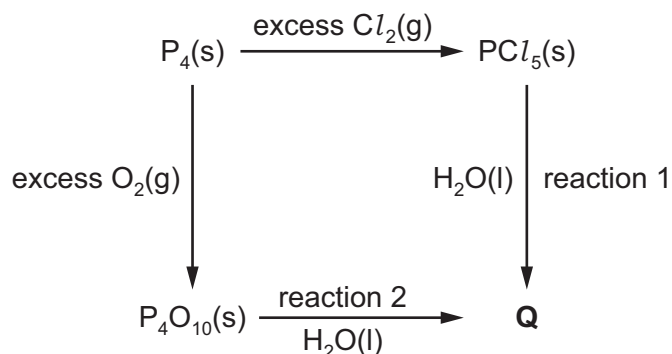
(ii) Reaction 3 is a redox reaction.

Identify which of the **reactants** is reduced in this reaction.

23. O/N 21/P23/Q3/b(i)

3 Phosphorus is a reactive Period 3 element.

(b) Some reactions of $P_4(s)$ are shown in the reaction scheme.

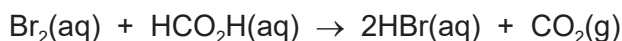


(i) State the oxidation number of phosphorus in P_4O_{10} .

[1]

24. M/J 21/P21/Q4/a

4 Aqueous bromine reacts with methanoic acid to form hydrogen bromide and carbon dioxide gas.



The table shows the oxidation numbers of bromine and carbon in the species involved in this reaction.

	Br in Br_2	C in HCO_2H	Br in HBr	C in CO_2
oxidation number	0	+2	-1	+4

(a) Identify the oxidising agent in this reaction. Explain your reasoning with reference to oxidation numbers.

[1]

25. M/J 21/P22/Q1/d(ii)

1 A Group 2 metal combines with bromine to form a crystalline solid, MBr_2 .

Excess aqueous $AgNO_3$ is added to a solution of MBr_2 and a precipitate forms. The mixture is filtered. The precipitate is dried and the mass of the precipitate is recorded.

(d) A sample of MBr_2 is dissolved in water. Chlorine gas is then bubbled into the solution.

(ii) Name the type of reaction that occurs when MBr_2 reacts with chlorine gas.

[1]

26. M/J 21/P23/Q1/f

1 The elements in Group 17 are known as the halogens.

- (f) Sodium chlorate(I), NaClO , oxidises dilute hydrochloric acid to form three products. The products which contain chlorine have chlorine species with oxidation number -1 or 0 .

No other species changes its oxidation number during the reaction.

Use this information to complete the ionic equation.

**27. MAR 21/P22/Q2/c(ii),d(i)**

2 Chlorine, Cl_2 , is a reactive yellow-green gas. It is a strong oxidising agent.

- (c) The halide ions, X^- (where $\text{X} = \text{Cl}, \text{Br}, \text{I}$), show clear trends in their physical and chemical properties.

- (ii) Complete the table to show the oxidation number of sulfur in each of the sulfur-containing products. [1]

- (d) Cl_2 reacts with aqueous sodium hydroxide in a disproportionation reaction.

- (i) State what is meant by *disproportionation*.

[1]

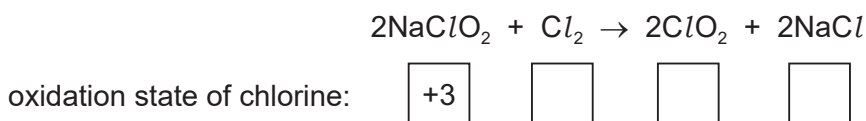
28. O/N 20/P21/Q2/d(ii)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

- (d) Chlorine forms several oxides, including Cl_2O , ClO_2 and Cl_2O_6 .

- (ii) ClO_2 can be prepared by reacting NaClO_2 with Cl_2 .

Write the oxidation state of chlorine in each species in the boxes provided.



[1]

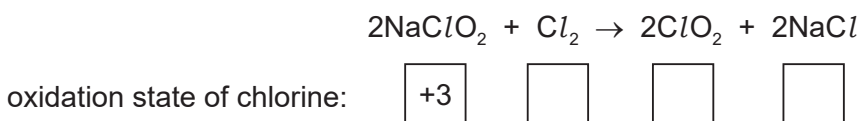
29. O/N 20/P23/Q2/d(ii)

2 Phosphorus, sulfur and chlorine can all react with oxygen to form oxides.

- (d) Chlorine forms several oxides, including Cl_2O , ClO_2 and Cl_2O_6 .

- (ii) ClO_2 can be prepared by reacting NaClO_2 with Cl_2 .

Write the oxidation state of chlorine in each species in the boxes provided.

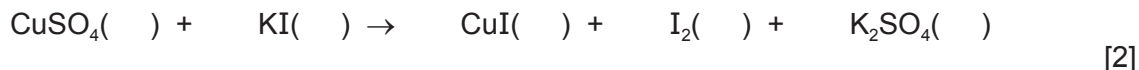


[1]

30. M/J 20/P21/Q2/a

- 2 (a) The equation shown in (a)(i) describes the reaction which occurs when aqueous potassium iodide is added to aqueous copper(II) sulfate. A white precipitate of copper(I) iodide forms in a brown solution of iodine and potassium sulfate.

(i) Balance the equation and include state symbols.



The table gives the oxidation numbers of iodine in the different species in the equation.

iodine-containing species	oxidation number of iodine
KI	−1
CuI	−1
I ₂	0

(ii) Deduce the oxidation number of copper in CuSO₄ and CuI.

- oxidation number of copper in CuSO₄
- oxidation number of copper in CuI

[1]

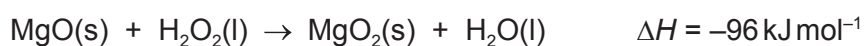
(iii) Describe the type of reaction shown by the equation in (a)(i). Explain your answer in terms of electron transfer.

[2]

31. MAR 20/P22/Q1/g(i)

- 1 Group 2 metals form alkaline solutions in water.

(g) Magnesium peroxide, MgO₂, is made in the following reaction.



compound	enthalpy change of formation, $\Delta H_f^\circ / \text{kJ mol}^{-1}$
MgO(s)	−602
H ₂ O ₂ (l)	−188
H ₂ O(l)	−286

(i) The peroxide ion is O₂^{2−}.

Deduce the average oxidation number of oxygen in the peroxide ion.

[1]

32. O/N 19/P21/Q1/a

- 1 (a) Chlorine can be prepared using the following reaction.



- (i) Explain why $\text{MnO}_2(\text{s})$ is described as an oxidising agent in this reaction.

Refer to oxidation numbers in your answer.

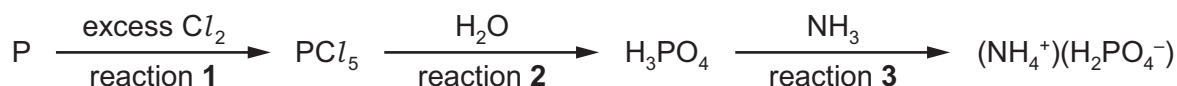
[1]

- (ii) State what you would observe during this reaction.

[1]

33. O/N 19/P22/Q3/a(iii)

- 3 A series of reactions for phosphorus and its compounds is shown.



- (iii) H_3PO_4 can be produced by direct reaction of phosphorus with nitric acid.



Use oxidation numbers to show that this reaction is a redox reaction.

[2]

34. O/N 19/P23/Q1/a

- 1 (a) Chlorine can be prepared using the following reaction.



- (i) Explain why $\text{MnO}_2(\text{s})$ is described as an oxidising agent in this reaction.

Refer to oxidation numbers in your answer.

[1]

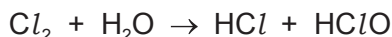
- (ii) State what you would observe during this reaction.

[1]

35. M/J 19/P22/Q3/b(i,ii)

3 Period 3 elements react with chlorine gas, $\text{Cl}_2(\text{g})$, to form chlorides.

(b) $\text{Cl}_2(\text{g})$ dissolves in cold water and reacts with it.



(i) Identify the oxidation number of chlorine in each of the chlorine-containing species in this reaction.

chlorine-containing species	Cl_2	HCl	HClO
oxidation number of chlorine			

[2]

(ii) Name the type of reaction occurring.

[1]

36. M/J 19/P23/Q2/c

2 Iodine is an element in Group 17 of the Periodic Table.

(c) In a reaction between hydrogen iodide and concentrated sulfuric acid, the products are hydrogen sulfide, sulfur, iodine and water.

(i) Write an equation for this reaction.

You may wish to use oxidation numbers to help you.

[2]

(ii) Explain, with reference to oxidation numbers, why this reaction is a redox reaction.

[2]

37. MAR 19/P22/Q1/d(i)

1 Nitrogen, N_2 , is the most abundant gas in the Earth's atmosphere and is very unreactive.

(d) Three oxides of nitrogen, NO , NO_2 and N_2O , can be formed under different conditions.

(i) Complete the table to give the oxidation numbers of nitrogen in NO and NO_2 .

compound	NO	NO_2
oxidation number of N		

[1]

38. O/N 18/P21/Q1/a(ii)

1 Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

(ii) Calculate the oxidation number of sulfur in the S_2^{2-} ion.

Assume that each sulfur atom in the ion has the same oxidation number.

oxidation number of sulfur in the S_2^{2-} ion =

[1]

39. O/N 18/P23/Q1/a(ii)

- 1 Iron pyrite, FeS_2 , has a yellow colour that makes it look like gold metal. The compound contains the ions Fe^{2+} and S_2^{2-} .

- (ii) Calculate the oxidation number of sulfur in the S_2^{2-} ion.
Assume that each sulfur atom in the ion has the same oxidation number.

oxidation number of sulfur in the S_2^{2-} ion = [1]

40. M/J 18/P22/Q2/d(i,ii)

- 2 Ammonium iron(II) sulfate, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2$, has a relative formula mass, M_r , of 284.

- (d) Mohr's salt, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot x\text{H}_2\text{O}$, is the hydrated form of ammonium iron(II) sulfate.

x represents the number of moles of water in 1 mole of the salt.

A student wanted to determine the value of x. 0.784 g of the hydrated salt was dissolved in water and this solution was acidified.

All of the solution was titrated with $0.0200 \text{ mol dm}^{-3}$ potassium manganate(VII). 20.0 cm^3 of this potassium manganate(VII) solution was required for complete reaction with the Fe^{2+} ions.

- (i) Use changes in oxidation numbers to balance the equation for the reaction taking place.



- (ii) State the role of the Fe^{2+} ions in this reaction.

Explain your answer.

[2]

41. M/J 18/P23/Q3/e(i)

- 3 The elements in Group 17, the halogens, show trends in both their chemical and physical properties. The elements and their compounds have a wide variety of uses.

- (e) Chlorine is commonly used in water purification. When chlorine is added to water it reacts to produce a mixture of acids, one of which is chloric(I) acid, HClO , a powerful oxidising agent.

- (i) Explain the meaning of the term *oxidising agent*, in terms of electron transfer.

[1]

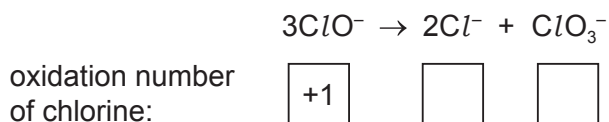
42. MAR 18/P22/Q3/c(iii,iv)

3 Calcium and its compounds have a large variety of applications.

(c) Calcium chlorate(I), $\text{Ca}(\text{ClO})_2$, is used as an alternative to sodium chlorate(I), NaClO , in some household products.

(iii) The chlorate(I) ion is unstable and decomposes when heated as shown.

Deduce the oxidation number of chlorine in each species. Complete the boxes.



[1]

(iv) In terms of electron transfer, state what happens to the chlorine in the reaction in **(iii)**.

[1]

43. M/J 16/P22/Q3/a

3 Acidified potassium dichromate(VI) can oxidise ethanedioic acid, $\text{H}_2\text{C}_2\text{O}_4$. The relevant half-equations are shown.



(a) State the overall equation for the reaction between acidified dichromate(VI) ions and ethanedioic acid.

[2]

44. M/J 16/P23/Q1/a(ii)

1 An experiment was carried out to determine the percentage of iron in a sample of iron wire.

(a) A 3.35 g piece of the wire was reacted with dilute sulfuric acid, in the absence of air, so that all of the iron atoms were converted to iron(II) ions. The resulting solution was made up to 250 cm^3 .

(ii) Use the half-equations to write an equation for the reaction between the iron(II) ions and the acidified dichromate(VI) ions.

[1]

45. M/J 16/P23/Q2/c

2 Ammonium nitrate is an important fertiliser made by the acid-base reaction between ammonia and nitric acid.

(c) The first stage in the production of nitric acid involves the reaction of ammonia with oxygen to form nitrogen monoxide, NO , and water.

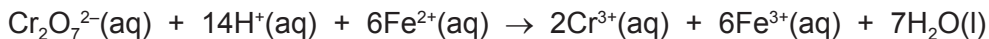
Suggest an equation for this reaction and use oxidation numbers to show that it is a redox reaction.

[3]

46. MAR 16/P22/Q2/b

- 2 Spathose is an iron ore that contains iron(II) carbonate, FeCO_3 . The percentage of iron(II) carbonate in spathose can be determined by titration with acidified potassium dichromate(VI) solution using a suitable indicator.

The ionic equation is shown below.



- (b) Iron ores containing iron(III) compounds can be analysed using a similar method.

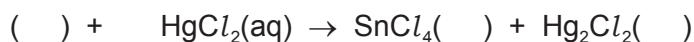
A standard solution of an aqueous iron(III) compound is reacted with aqueous tin(II) chloride. Aqueous tin(IV) chloride and aqueous iron(II) chloride are the products of this reaction.

- (i) Write an **ionic** equation for this reaction. Do not include state symbols.

[2]

- (ii) Any excess tin(II) chloride can be removed by reaction with $\text{HgCl}_2(\text{aq})$. A white precipitate of Hg_2Cl_2 is produced.

Complete the equation for this reaction.



[2]

47. M/J 15/P22/Q2/a(ii)

- 2 Sulfuric acid is an important chemical with a variety of uses.

It is manufactured by the Contact process, the first stage of which involves the conversion of sulfur or a sulfide ore, such as galena, PbS , into sulfur dioxide, SO_2 .

- (ii) Identify the oxidation number changes that take place during this reaction.

[2]

Answers Section

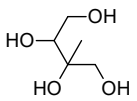
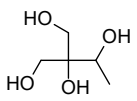
1. M/J 26/P21/Q3/d

3(d)	species	oxidation number of sulfur	2 •✓✓ A 6 (+) 4 (+) and 2– A VI (+) AND IV (+) AND II – R S ⁺⁴ S ⁺⁶ or S ^{–2} R S ⁴⁺ S ⁶⁺ or S ^{2–}
	H ₂ SO ₄	(+)VI OR (+)6	
	SO ₂	(+)IV OR (+)4	
	H ₂ S	–II OR –2	

2. M/J 26/P21/Q4/a(ii)

4(a)(ii)	$\text{C}_5\text{H}_{12}\text{O}_4 + 2\text{Cr}_2\text{O}_7^{2-} + 16\text{H}^+ \rightarrow \text{C}_5\text{H}_6\text{O}_7 + 4\text{Cr}^{3+} + 11\text{H}_2\text{O}$	1	
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3. M/J 26/P22/Q4/c

4(c)(i)	$3\text{C}_5\text{H}_{12}\text{O}_4 + 5\text{Cr}_2\text{O}_7^{2-} + 40\text{H}^+ \rightarrow 3\text{C}_5\text{H}_6\text{O}_8 + 10\text{Cr}^{3+} + 29\text{H}_2\text{O}$ M1 $5\text{Cr}_2\text{O}_7^{2-}$ M2 fully correct equation	2	
4(c)(ii)	HOCH ₂ CH(OH)C(OH)(CH ₃)CH ₂ OH OR (HOCH ₂) ₂ C(OH)(CHOHCH ₃)  OR  • molecular formula C₅H₁₂O₄ • 2 primary alcohol groups • 1 secondary alcohol group • 1 tertiary alcohol group •✓•✓	2	A unambiguous structure I any name attempted Max 1 all correct but 1 or more connectivity issue e.g. C–H–O

4. M/J 26/P23/Q4/d(ii)

4(d)(ii)	$3\text{C}_5\text{H}_{12}\text{O}_4 + 8\text{Cr}_2\text{O}_7^{2-} + 64\text{H}^+ \rightarrow 3\text{C}_5\text{H}_4\text{O}_8 + 16\text{Cr}^{3+} + 44\text{H}_2\text{O}$	1	
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5. M/J 25/P21/Q4/b(iii)

4(b)(iii)	oxidising agent	1
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6. M/J 25/P24/Q2/b(ii)

2(b)(ii)	oxidation number of S in $\text{SCl}_2 = (+)2 / (+)\text{II}$ oxidation number of S in $\text{H}_2\text{SO}_3 = (+)4 / (+)\text{IV}$	2
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7. MAR 25/P22/Q1/a

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	2
1(a)(ii)	high pressure low temperature	2
1(a)(iii)	ratio of moles of P : Cl = $\frac{30.4}{31.0} : \frac{69.6}{35.5} = 0.98 : 1.96 (= 1 : 2)$ molecular formula = P_2Cl_4	2
1(a)(iv)	(structure =) simple / molecular because low melting point / weak intermolecular forces (bonding =) covalent because hydrolysed	2

8. O/N 24/P21/Q2/c(ii,iii)

2(c)(ii)	a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction	1
2(c)(iii)	$\text{CH}_3\text{COCH}_3 + 3\text{NaClO} \rightarrow 1\text{CHCl}_3 + 1\text{CH}_3\text{COONa} + 2\text{NaOH}$	1

9. O/N 24/P22/Q3/b(i,ii)

3(b)(i)	(+)5 / V	1
3(b)(ii)	aluminium / Al	1

10. O/N 24/P23/Q2/c(ii,iii)

2(c)(ii)	a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction	1
2(c)(iii)	$\text{CH}_3\text{COCH}_3 + 3\text{NaClO} \rightarrow 1\text{CHCl}_3 + 1\text{CH}_3\text{COONa} + 2\text{NaOH}$	1

11. M/J 24/P23/Q1/d(i)

1(d)(i)	(+)IV	1
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12. MAR 24/P22/Q1/b(i),c,d(i)

1(b)(i)	$2\text{Bi} + 3\text{H}_2\text{O} \rightarrow \text{Bi}_2\text{O}_3 + 3\text{H}_2$	1
1(c)(i)	Bi_2O_3 (+)3 / III AND NaBiO_3 (+)5 / V	1
1(c)(ii)	Bi_2O_3	1
1(d)(i)	a substance that removes electrons from another / causes electron loss / gains electrons	1

13. O/N 23/P21/Q3/a,c

3(a)	compound / molecule / substance that oxidises another / is reduced	1
3(c)	Cl / chlorine is both oxidised and reduced	1

14. O/N 23/P22/Q3/b(iii)

3(b)(iii)	P (in H_3PO_3) is (both) oxidised and reduced (simultaneously)	1
	P (is oxidised) from (+)3 / (+)III $\rightarrow$ (+)5 / (+)V AND (reduced to) -3 / -III	1

15. O/N 23/P23/Q3/a,c

3(a)	compound / molecule / substance that oxidises another / is reduced	1
3(c)	Cl / chlorine is both oxidised and reduced	1

16. M/J 23/P22/Q2/e(ii)

2(e)(ii)	$\text{NaClO} + \text{H}_2\text{O}_2 \rightarrow \text{O}_2 + \text{NaCl} + \text{H}_2\text{O}$	1
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17. M/J 23/P23/Q1/d(i,ii)

1(d)(i)	$2\text{Cu}^{2+} + 4\text{I}^- \rightarrow 2\text{CuI} + \text{I}_2$	1
1(d)(ii)	Cu^{2+} OR CuSO_4 AND copper (species / ion) has gained / taken electron(s) from iodide (ion)	1

18. M/J 22/P22/Q1/b(i,ii)

1(b)(i)	species	magnesium in Mg_3N_2	nitrogen in Mg_3N_2	1
	oxidation number	2(+)/(+)II	3−/−III	
1(b)(ii)	- redox - Mg loses electrons / Mg increases ox no / Mg is oxidised - N gains electrons / N reduces ox no / N is reduced			1

19. MAR 22/P22/Q2/a(i),c(ii)

2(a)(i)	species that donates electrons	1
2(c)(ii)	(+)3 / III	1

20. O/N 21/P21/Q3/b(i)

3(b)(i)	(+)5/V	1
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21. O/N 21/P22/Q1/d

1(d)(i)	$4\text{HI} + \text{O}_2 \rightarrow 2\text{I}_2 + 2\text{H}_2\text{O}$	1
1(d)(ii)	M1 I / iodine (increases) oxidation number $-1 \rightarrow 0$ so oxidation / is oxidised OR HI / is oxidised as I (increases) oxidation number $-1 \rightarrow 0$ M2 O (decreases) oxidation number $0 \rightarrow -2$ so reduction / is reduced	2

22. O/N 21/P22/Q3/e(ii)

3(e)(ii)	Iodine / I_2	1
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23. O/N 21/P23/Q3/b(i)

3(b)(i)	(+)5/V	1
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24. M/J 21/P21/Q4/a

4(a)	Br / bromine as the oxidation number of Br decreases / goes from $0 \rightarrow -1$ OR bromine as it causes oxidation number of C (in methanoic acid) to increase / go from $(+)2 \rightarrow (+)4$	1
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25. M/J 21/P22/Q1/d(ii)

1(d)(ii)	displacement	1
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26. M/J 21/P23/Q1/f

1(f)	M1 products contain Cl^- AND Cl_2	1
	M2 correct balanced equation $\text{ClO}^- + 2\text{HCl} \rightarrow \text{Cl}^- + \text{H}_2\text{O} + \text{Cl}_2$	1

27. MAR 21/P22/Q2/c(ii),d(i)

2(c)(ii)	(+)6, (+)4, -2	1
2(d)(i)	when a species is both oxidised and reduced	1

28. O/N 20/P21/Q2/d(ii)

2(d)(ii)	0 (+4) -1	1
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29. O/N 20/P23/Q2/d(ii)

2(d)(ii)	0 (+)4 -1	1
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30. M/J 20/P21/Q2/a

2(a)(i)	$2\text{CuSO}_4(\text{aq}) + 4\text{KI}(\text{aq}) \rightarrow 2\text{CuI}(\text{s}) + (1)\text{I}_2(\text{aq}) + 2\text{K}_2\text{SO}_4(\text{aq})$ M1 correct balancing M2 correct state symbols	2
2(a)(ii)	Oxidation state of copper in CuSO_4 (+)2 AND Oxidation state of copper in CuI (+)1	1
2(a)(iii)	M1 redox	1
	M2 iodide ions – lost electron(s) AND copper ions – gained electron(s)	1

31. MAR 20/P22/Q1/g(i)

1(g)(i)	-1	1
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32. O/N 19/P21/Q1/a

1(a)(ii)	It oxidises chlorine from -1 to 0	1
1(a)(ii)	effervescence / fizzing / bubbling OR <u>green</u> gas formed OR solid dissolves / disappears / soluble	1

33. O/N 19/P22/Q3/a(iii)

3(a)(iii)	- P goes from 0 to (+)5 / (+)V - P is oxidised - N goes from (+)5 / (+)V to (+)4 / (+)IV - N is reduced Award one mark for two correct bullet points, award two marks for all four correct.	2
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34. O/N 19/P23/Q1/a

1(a)(ii)	It oxidises chlorine from -1 to 0	1
1(a)(ii)	effervescence / fizzing / bubbling OR <u>green</u> gas formed OR solid dissolves / disappears / soluble	1

35. M/J 19/P22/Q3/b(i,ii)

3(b)(i)	<table><tr><td>Chlorine containing species</td><td>Cl_2</td><td>HCl</td><td>HOCl</td></tr><tr><td>Oxidation number of chlorine</td><td>0</td><td>-1</td><td>(+)1</td></tr></table>				Chlorine containing species	Cl_2	HCl	HOCl	Oxidation number of chlorine	0	-1	(+)1	2
	Chlorine containing species	Cl_2	HCl	HOCl									
	Oxidation number of chlorine	0	-1	(+)1									
Award 2 marks for 3 correct oxidation numbers													
Award 1 Mark for 2 correct oxidation numbers													
3(b)(ii)	disproportionation				1								

36. M/J 19/P23/Q2/c

2(c)(i)	$2\text{H}_2\text{SO}_4 + 14\text{HI} \rightarrow 7\text{I}_2 + 8\text{H}_2\text{O} + \text{H}_2\text{S} + \text{S}$ M1 correct species M2 correctly balanced equation	2
2(c)(ii)	<i>explain with ref to ox no's why the reaction in (c)(i) is a redox reaction</i> M1 I (oxidation number increases) from -1 to 0 = oxidation / reducing agent M2 S (oxidation number decreases) from (+) 6 to 0 OR -2 = reduction / oxidising agent	2

37. MAR 19/P22/Q1/d(i)

1(d)(i)		NO	NO ₂	
		(+)2 / (+)II	(+)4 / (+)IV	

38. O/N 18/P21/Q1/a(ii)

1(a)(ii)	-1	1
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39. O/N 18/P23/Q1/a(ii)

1(a)(ii)	-1	1
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40. M/J 18/P22/Q2/d(i,ii)

2(d)(i)	$\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$	1
2(d)(ii)	(Fe ²⁺ is a) reducing agent / reductant provides/donates electron(s) / loses electron(s) / increases its oxidation number / (Fe ²⁺) becomes Fe ³⁺	2

41. M/J 18/P23/Q3/e(i)

3(e)(i)	(Species that) gains <u>electrons</u> / <u>electron</u> acceptor	1
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42. MAR 18/P22/Q3/c(iii,iv)

3(c)(iii)	-1 AND (+)5	1
3(c)(iv)	gains AND loses electrons	1

43. M/J 16/P22/Q3/a

3 (a)	$\text{Cr}_2\text{O}_7^{2-} + 8\text{H}^+ + 3\text{H}_2\text{C}_2\text{O}_4 \rightarrow 2\text{Cr}^{3+} + 6\text{CO}_2 + 7\text{H}_2\text{O}$ M1 = species M2 = balancing	[1] [1]	[2]
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44. M/J 16/P23/Q1/a(ii)

(ii)	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{Fe}^{2+} \rightarrow 2\text{Cr}^{3+} + 6\text{Fe}^{3+} + 7\text{H}_2\text{O}$	[1]	[1]
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45. M/J 16/P23/Q2/c

(c)	$4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$ N changes from -3 to +2 (so oxidation) O changes from 0 to -2 (so reduction)	[1] [1] [1]	[3]
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46. MAR 16/P22/Q2/b

(b) (i)	$2\text{Fe}^{3+} + \text{Sn}^{2+} \rightarrow 2\text{Fe}^{2+} + \text{Sn}^{4+}$ species balancing	[1] [1]	[2]
(ii)	$\text{SnCl}_2(\text{aq}) + 2\text{HgCl}_2(\text{aq}) \rightarrow \text{SnCl}_4(\text{aq}) + \text{Hg}_2\text{Cl}_2(\text{s})$ SnCl ₂ AND 2 state symbols	[1] [1]	[2]

47. M/J 15/P22/Q2/a(ii)

(ii)	S (is oxidised) -2 to (+)4 O (is reduced) 0 to -2	[1] [1]	[2]
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